smar

FIRST IN FIELDBUS



The Fieldbus Reference Book

FIRST IN FIELDBUS

RST IN FIELDBUS

Exploring the Future



Mr. Jonas Berge the general manager of Smar Singapore wrote has written a comprehensive 500-page book about Fieldbus Technology for the Instrument Society of America. I asked him for a signed copy.

I was pleased to receive it with a very nice dedicatory. He wrote:

"It's a pleasure exploring the future with you".

I then began to reflect over the past years and realized how right he is. We at Smar are always exploring the future. Today we are designing products that will be dominant in the market over the next years. This strategy has proved right as our organic growth shows. While most of the companies in the instrumentation business are stable or growing through mergers and acquisitions, Smar is growing the healthy old fashioned way, through revenues and profits, based on growing sales.

Exploring the future can bring high prizes if you are right. Losses can be immense if you are wrong. It's not an easy task. For one single good reason: The future is not for us to see. A lot of research, guessing, intuition and even good luck is required. Fortunately we have managed to have all of that together over the years.

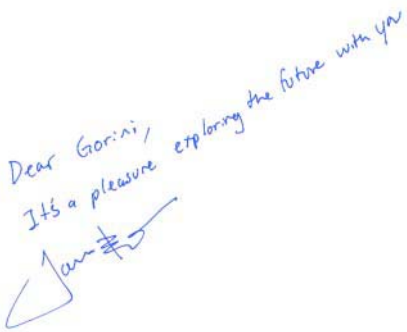
10 years ago we envisioned that Fieldbus would be the future of process control and started designing Fieldbus products and solutions. Today we see that more and more customers are already buying Fieldbus and this technology will shortly be the preferred choice for customers all around the world. Smar technology in this area is clearly recognized as the best in the world.

2001 was a very good year for Smar. In spite of all of the big disturbances, crisis and recession in the world, sales went up 34% and profits went up 69 %. Because of Fieldbus and the successful installation of several System302s integrated solutions in the market, Smar becomes more and more the reference for open control system solutions, based on Fieldbus. The momentum is still very high and we expect very good results for the next years.

More and more we are expanding our R&D activities to better explore the future bringing into the market the products that will control the world in the coming years.



Jonas with his book



Edmundo Gorini
President

Fieldbus Advantages

True integration of all components

Fieldbus keeps the signal purely digital all the way, to the end of the chain, avoiding loss of accuracy or signal integrity.

Everything interoperable

Multiple devices, independent of the manufacturer, will operate in the same system, without any loss of functionality.

True field control

The control strategy is distributed among the field devices, ensuring greater fault tolerance and availability.

Device Maintenance

Powerful proactive maintenance and reporting information, based on the increased diagnostic data available from the field devices, reduces unnecessary field trips.

Simplicity of operation

From any workstation the operator can configure the entire control strategy and devices in the field.

Cost Savings

Fieldbus provides lower investments in equipment and in installation, reduces maintenance cost, in addition to reduced engineering costs due to simplified configuration of control strategies.

Customer Satisfaction

"The 'intelligent' instrumentation not only saved the company money, but took less time to install and configure. We are extremely pleased with the outcome of this project. It has exceeded all our expectations." Smar Notes February/March, 1997.

Mr. Philippe Mille,

Project Engineer - Societé Française de Thermolyse (SFT)

"The main advantages of SYSTEM302 are its low implementation cost and the maintenance diagnostic. In addition to Smar's high quality of service. " Smar Notes December, 2001.

Mr. Hamilton Baldo,

Senior Instrumentation Technician - Rhodia





"The reason we want to buy Smar Fieldbus solution is because Smar is the only solution provider, among others, with real working Fieldbus systems based on its own technology." Smar Notes June/July, 1997.

Mr. Zhang Guangchuan,

President - Beijing Huakong Technology Co., Ltd.

"Fieldbus is cost-effective, simple, powerful, reliable, and a lot more. A traditional DCS system simply loses out to the virtues of a Fieldbus systems in nearly all aspects." Smar Notes June/July, 1997.

Mr. Niu Qi,

Vice-President - Binzhou Chemical Company



"The time for installation and start up was shortest than conventional systems due to the simplicity of wiring offered by twisted pair bus and also by the ability to calibrate and configure devices from the workstation. In terms of operation I can say Fieldbus is high reliable system." - Smar Notes December, 1997.

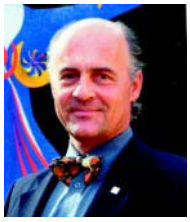
Mr. Rogelio Berber Guzman,

Chief of Instrumentation dept. - CFE CT José Aceves Pozos

"The Fieldbus System installed by Smar in our factory at Camaçari - Bahia is a complete success. In addition to reducing our operational losses and maintenance costs, it significantly contributed to the improvement of our product quality. Regarding our productivity, Smar Fieldbus has given our personnel a great professional motivation, making it possible for us to put in operation the Advanced Process Control and have other gains.

Eng. Luiz Petersen,

Instrumentation and Electrical Coordinator - DETEN



"The great experience that SMAR has in Foundation Fieldbus in particular will strengthen the range of process automation solutions that we can offer to our customers world wide." Smar Notes December, 2001.

Mr. Klaus Endress,

CEO and Group's chief operating officer - Endress + Hauser

"We know Smar is far ahead of others in Fieldbus technology, and we know Smar's Fieldbus Control Systems work, and work much better than conventional systems." - Smar Notes June/July, 1997.

Chief Instrumentation Engineer

The Chemical Ministry Design Institute

"We choose SYSTEM302 to carry through the automation of this important plant for the environment, because its devices are of latest generation and they provide a safer operation and a tight control to achieve a end product with better quality." - Smar Notes July, 2001.

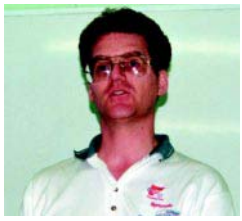
Mr. Maurício Mascolo,

Process Engineer - Univen.

Smar has always been and continues to be "First in Fieldbus." Not only in support of but also in the development of this technology. Smar has proven this to me many times since I have started using their equipment in 1996, that they have an exceptional group of engineers who quickly develop concepts into quality products and solutions..

Ian Verhappen,

Fieldbus User Network





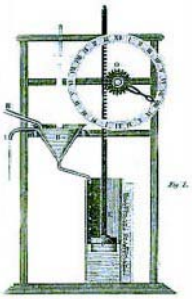
"The Fieldbus System version ISP-50 was a pioneer at DETEN, as it also pioneered in the world. In spite of the difficulties involving the implementation of an entirely new system, it was successfully carried out with a lot of zeal and a lot of work by DETEN and SMAR partnership.
Its performance, reliability and operational continuity are excellent, beyond expectations, and we no longer have problems such as freezing, communication errors, loss of information, etc.
The ISP-50 reduced our operational loss and maintenance costs. It was surely the right investment and made us very pleased!"

Alexandre Pessoa,

Instrumentation Engineer - DETEN

Fieldbus:

The revolutionary technology that is changing the world of automation.

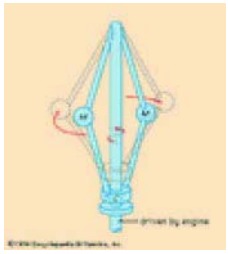


270 BC: Float regulator for a water clock.

1588: Invention of the mill-hopper.

1624: Automatic temperature control system for a furnace.

1745: Fantail for a windmill.



1774: Flyball governor for steam engine applications.

1840: Feedback device for pointing a telescope.

1883: Invention of the damper-flapper regulator.



1888: Constant pressure pump governor.

1928: Pneumatic operational amplifier.

2002:
Fieldbus Age



1994: First Fieldbus Plant.



2000: First registered HSE device.

1992: First Fieldbus Chip.





1987: Fieldbus Standardization Studies.

1975: First distributed digital control system.

1984: First PC based supervisory control and data acquisition system.



1969: First Programmable Logical Controller



1973: Electrically driven industrial robot.

1964: First automated boiler control system.

1969: Analog pressure transmitter.

1959: First supervisory digital computer in self-employed power plant and refinery application.

1948: First pneumatic differential - pressure transmitter.



Fieldbus brings to the industrial environments the same advantages and easiness of operation that computer networks brought to the offices in the recent years. Fieldbus is a digital network that interconnects field instruments, such as transmitters, controllers and actuators with each other and the host computer.

Advanced features in hardware and software make it robust, reliable, safe, powerful, elegant and easy to install and use.

First in Fieldbus

As the leading company in developing and applying Fieldbus technology Smar has achieved several firsts.

Here are some of the firsts from Smar…



FB1050

World's First Fieldbus Chip. (1992)

LD302

World's First Fieldbus Device to receive the Control Engineering Product Recognition Award (1994)





IF, FI e FP302

World's First Fieldbus / 4-20 mA / Pressure Converters (1995)

Deten

World's First commercial Fieldbus plant. (1994)





LC700

World's First Fieldbus Programmable Controller. (1995)

FY302

World's First Fieldbus Valve Positioner (1996)



FB Stack

World's First Fieldbus Stack approved by Fieldbus Foundation. (1997)

FOUNDATION™ HSE High Speed Ethernet

First Registered HSE device and commercial installation. (2000)

1994 to 2000

First Commercial Fieldbus Installation in the following countries:

Germany
Belgium
Canada
China
United States
Italy
Mexico
Jordan



Argentina
Brazil
Chile
Singapore
France
England
Norway

Thousands of applications in hundreds of installations.

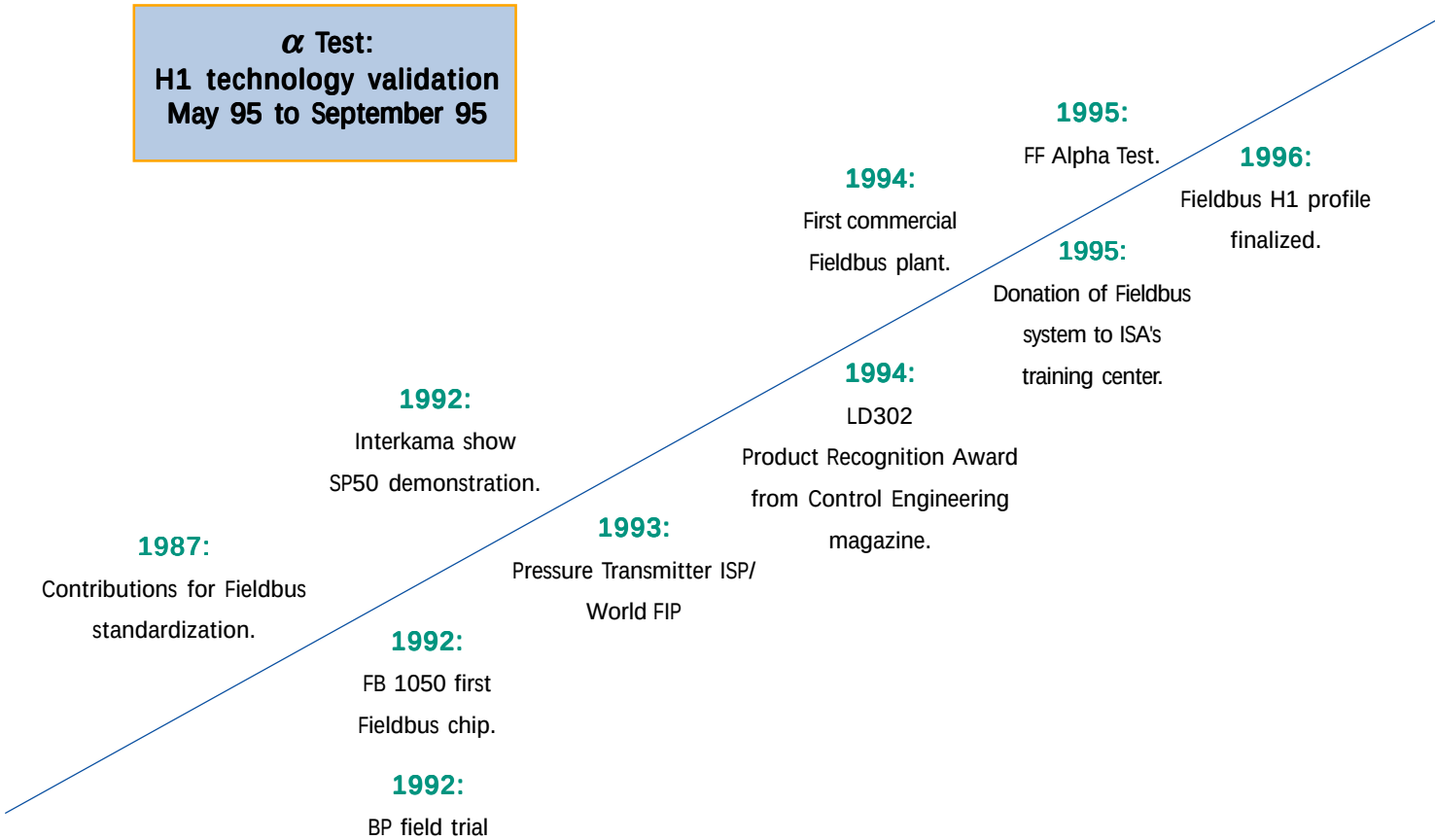
World's largest installed base.

Committed to the development of the open technology

Over the last 15 years Smar has significantly contributed to the development and standardization of the technology.

ß Test:
the Monsanto Co. Chocolate Bayou petrochemical complex
December 95 to April 96

α Test:
H1 technology validation
May 95 to September 95



1987:
Contributions for Fieldbus standardization.

1992:
Interkama show
SP50 demonstration.

1992:
FB 1050 first
Fieldbus chip.

1992:
BP field trial
IEC/SP 50 test.

1993:
Pressure Transmitter ISP/
World FIP

1994:
First commercial
Fieldbus plant.

1994:
LD302
Product Recognition Award
from Control Engineering
magazine.

1995:
FF Alpha Test.

1995:
Donation of Fieldbus
system to ISA's
training center.

1996:
Fieldbus H1 profile
finalized.

1997:
Communication Stack registered.

1997:
Daishowa - First Fieldbus interoperable production installation.

1997:
FY302 - Product Recognition Award from Control Engineering magazine.

1998:
SYSTEM302 version 4.0, a complete system for industrial automation.

1998:
FF Beta Test.

1998:
Syncrude Trial.

1998:
Sira Field Trial.

1999:
American Automatic Control Conference.

1999:
First Fieldbus Mobile Training Center.



2000:
DFI302 HSE linking device.

2001:
Smar Albert F. Sperry Award.

2001:
The First Book About Fieldbus "Fieldbuses for Process Control"

Fieldbus Trials



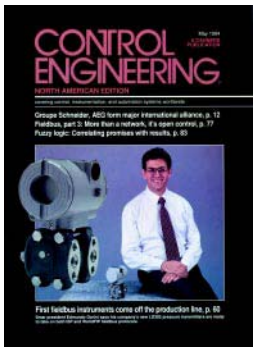
The Syncrude Trial occurred in 1998 to evaluate operating requirements and devices' communication with each other, a host or a distributed control system.



Smar provided technical support and equipment for Sira to set up a facility to provide practical demonstrations of process control products and systems that incorporate Fieldbus technology.

Awards an Recognition

Smar has received several awards and recognitions for the outstanding results in the development of the fieldbus technology.





1994:
LD302 Fieldbus Pressure Transmitter - Product Recognition Award from Control Engineering magazine.



1995:
Recognition for the significant time and resources devoted to bringing the advantages of ISA Membership to measurement and control professionals throughout South America and for the donation of Fieldbus instrumentation for use in ISA training programs, to Smar and its Management, on December.



1996:
Specification Team Leader for the Fieldbus Foundation
Mr. Libânio Carlos de Souza.

1996:
Pressure Profile Team Leader for the Fieldbus Foundation
Mr. Eugênio F. da Silva Neto.



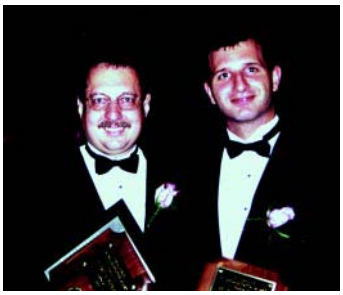
1996:
Excellence in Documentation Award - **"Three Views of Fieldbus in Water and Wastewater Applications"** by Libânio Carlos de Souza, Edward W. Baltutis, Mark T. Hood, and Joseph Signorelli.



1997:
FY302 Fieldbus Valve Positioner
Product Recognition Award from Control Engineering magazine.

1999:
Excellence in Documentation Award - **"Addressing Benefits and FAQ's of Fieldbus Based FCS Architecture"** by Jonas Berge.





2000:
Excellence in Documentation Award - **"Reliability with Foundation Fieldbus"** by Claudio Fayad and Pedro Biondo.

2000:
Data Link Bridge Team Member, on the High Speed Ethernet Development Program - Mr. Libânio Carlos de Souza.



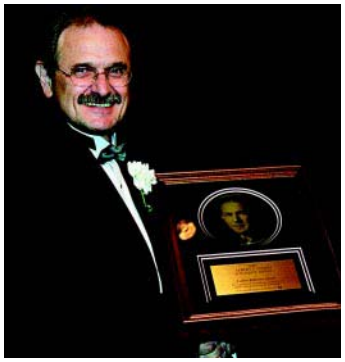


2001:
DT301 Density Transmitter
Control Engineering Editor's Choice.

2001:
To the success of the High Speed Ethernet Development Program - Smar Linking Device Development Team.



2000:
Multiple Input/Output (MIO) Team Leader on the High Speed Ethernet Development Program - Mr. Sérgio Hideki Tateishi.



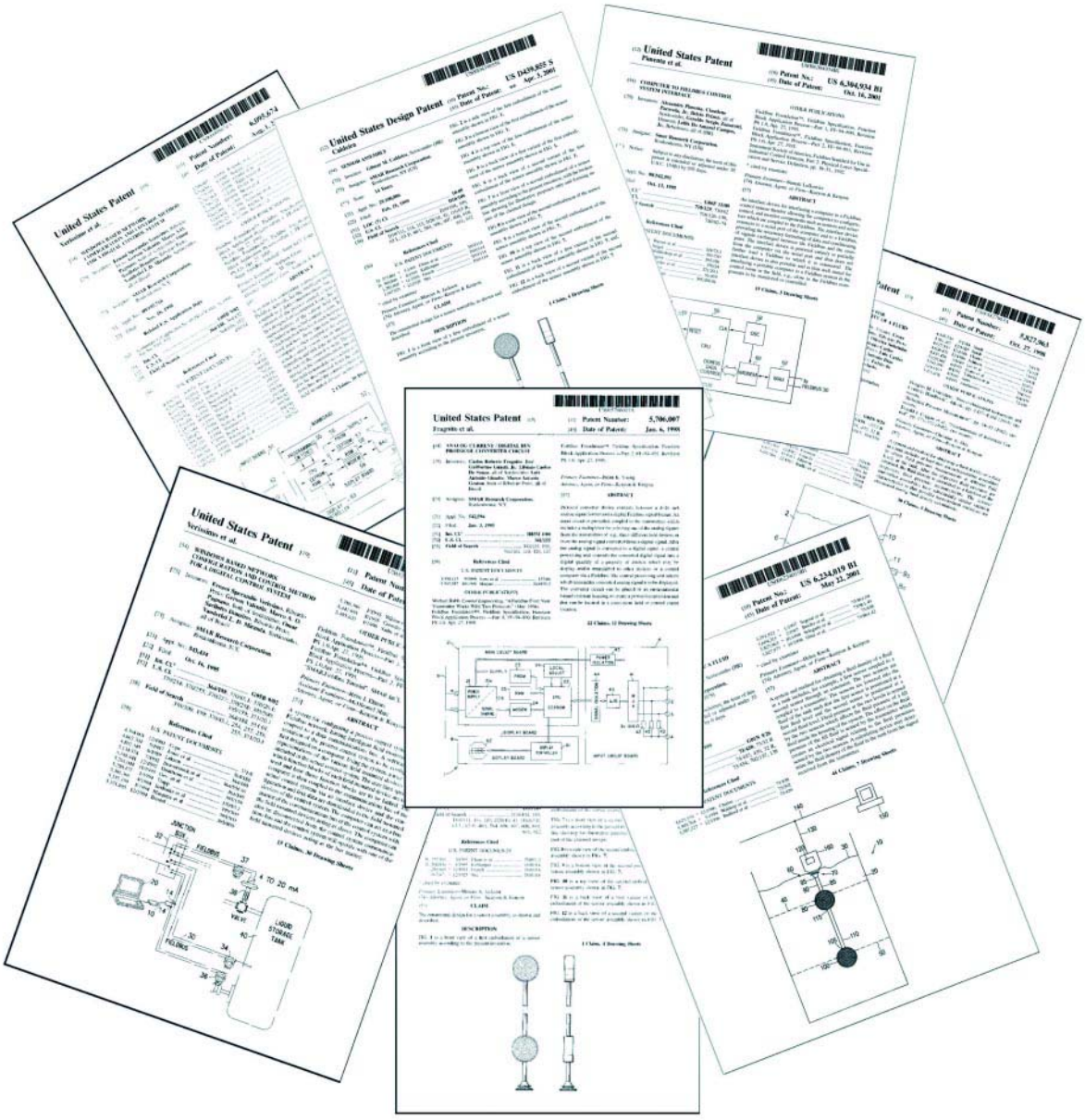
2001:
Recognition of Smar Dedication to the Fieldbus Technology, ISA Albert F. Sperry Award.



2001:
First registered device.

Patents

Smar has applied for and obtained several patents covering its products.

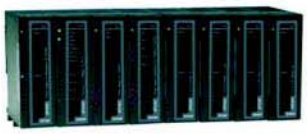


Foundation Fieldbus Products

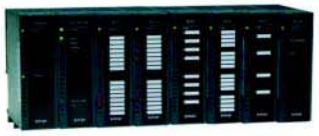
Smar has the world's largest line of Fieldbus products.



Workstations



DFI302
Universal Bridge



LC700
Programmable Controller



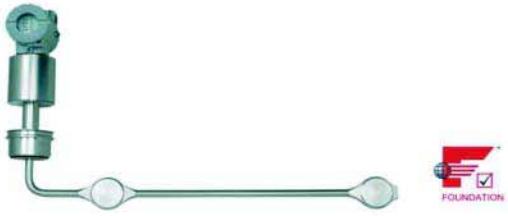
LD292
Pressure Transmitter



FI302
Fieldbus to Current Converter



FY302
Valve Positioner



DT302
Concentration/Density Transmitter



LD302
Pressure Transmitter



TP302
Position Transmitter



Asset View



Process View



Syscon



Smar OLE Server



Tag View



CONF700



BC302
Fieldbus/USB Interface



PCI
Process Control
Interface



DC302
Remote I/O



TT302
Temperature
Transmitter



FY402
Single Acting Valve
Positioner

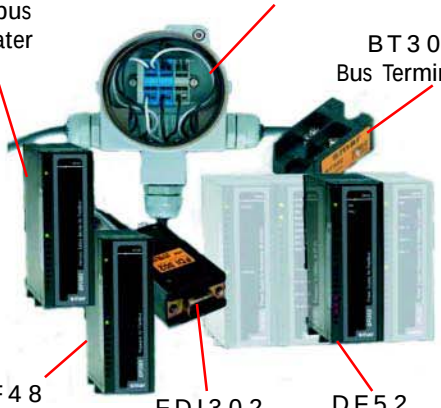


DF53 Power Supply

DF47
Intrinsically
Safe Barrier
and H1
Fieldbus
Repeater

JM1
Junction Box for

BT302
Bus Terminator





FP302
Fieldbus to Pneumatic
Signal Converter



IF302
Current to Fieldbus
Converter

DF48
H1 Fieldbus Repeater

FDI302
Field Device
Interface

DF52
Power Supply

Profibus Products



TT303
Profibus PA Temperature Transmitter



LD303
Profibus PA Pressure Transmitter



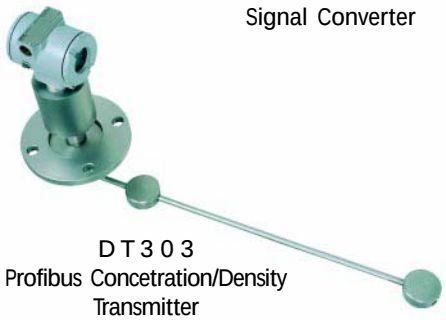
IF303
Current to Profibus PA Converter



FI303
Profibus PA to Current Converter



FP303
Profibus PA to Pneumatic Signal Converter

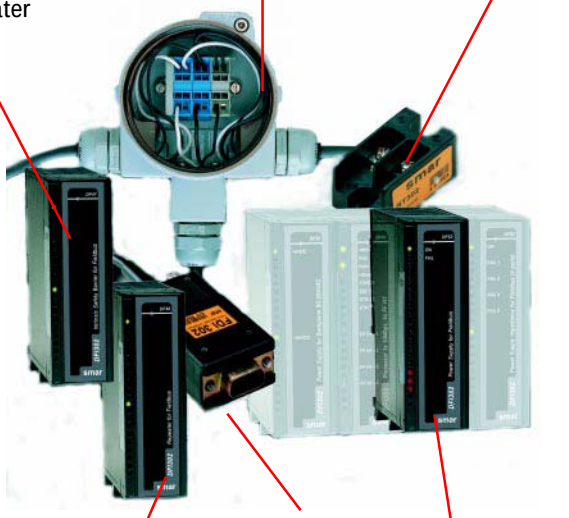


DT303
Profibus Concetration/Density Transmitter



FY303
Profibus PA Valve Positioner

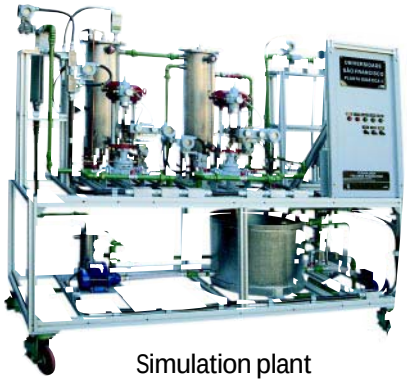
Other Smar products that can be used with 303 Series - Profibus PA



Training:

Since the beginning of its involvement with Fieldbus, Smar has been leading the promotion of this technology to users around the world.

Smar has donated equipment to several entities, Universities and Technical Schools, such as ISA, Lee College, SENAI and others.



Simulation plant



Demo-kit

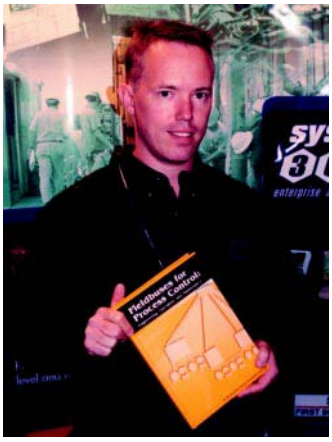




The world's first mobile training center for teaching and learning Fieldbus technology started operation in August 1999 in Brazil.

ISA Publishes the First Book About Fieldbuses

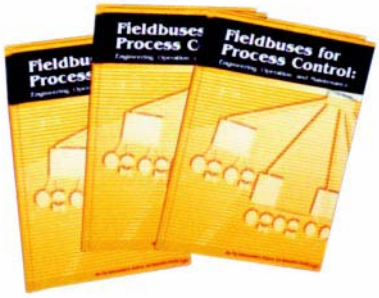
Finally the first book about fieldbuses has been published by ISA. Debuting at ISA 2001, Fieldbuses for Process Control, by Jonas Berge, Smar Singapore General Manager, is the result of years of research of the author in this field.



Rather than trying to cover all industrial communications, Jonas focused on the field-level process networks HART, FOUNDATION H1 Fieldbus, PROFIBUS PA and the host-level Ethernet versions FOUNDATION HSE Fieldbus and PROFInet. More than just wiring rules, this book covers the topic in-depth including configuration and documentation aspects. The book has 460 hardbound pages, approximately 230 illustrations and 100 review questions.

This implementation-oriented book based on experience collected from end-users across a wide range of industries around the world provides a clear and concise presentation of how to apply fieldbuses for process control. It addresses "how-to" for all phases of the system lifecycle from engineering to device and strategy configuration, installation and commissioning, diagnostics and troubleshooting, operation, calibration and maintenance. The book covers the three leading process fieldbus technologies: HART, FOUNDATION Fieldbus and PROFIBUS-PA. Both the field-level and the Ethernet-based host-level networking are covered.

The operation chapter touches on software integration using OPC (OLE for Process Control). The text also addresses concerns and solutions for interoperability, integration and migration as well as availability and safety. A chapter on benefits helps engineers justify business advantages to the management. The final chapter is for "propeller heads" and covers how these fieldbus technologies work. The book exposes similarities, differences and capabilities of these bus technologies.



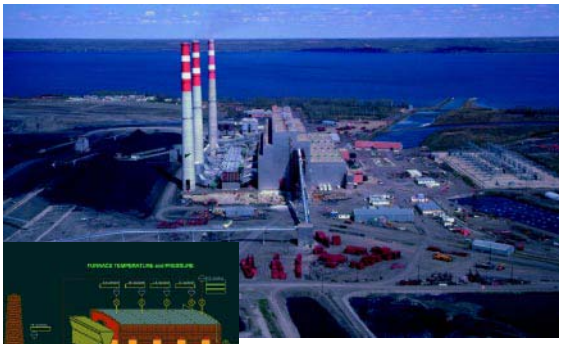
This book is a “must” for system designers, instrument engineers and technicians. Even process engineers can benefit learning about the capabilities of fieldbus technologies. The book is ideal for courses and for self-study and will remain a handy reference when configuring and troubleshooting systems. This book is sure to be a well-thumbed addition to every control engineer's bookshelf.

Published Papers

Smar technology is widely recognized in the press.	
Fieldbus - Improving your Measurement Digitally	ICARCV 1996
Fieldbus Enables Innovative Measurements	InTech 1997
Digital Fieldbus Technology Unleashes Full Potential of Intelligent Field Devices	I&CS 1997
Solução Fieldbus na Usina Santa Elisa S.A.	Petro & Química 1997
Fieldbus, its Benefits and Use	ISA-Japan 1997
Fieldbus In - DCS Out	Hydrocarbon Asia 1997
System 302 - Um Novo Horizonte para a Automação Industrial	RNA Automação 1998
Fieldbus Advances Diagnostics	InTech 1998
Integración de Sistemas Fieldbus Foundation con Sistemas de E/S Discretas y Enclavamiento	Instrumentación & Control Automatico 1998
For Remote Stations, Fieldbus + PLC + Radio = Economical Network	InTech 1999
Adaptive Control Using Fieldbus Resources	IsaTech 1999
Managing a Mixed DCS and Fieldbus System Environment	IsaTech 1999
New Engineering Practices and Tools for Fieldbus Based Systems	IsaTech 1999
Density Measurement: An Easy Solution Using Digital Transmitters	IsaTech 1999
New Protocol Widens Fieldbus	InTech 1999
Convenient Smart and Fieldbus Positioners Reduce Cost and Effort	IsaTech 1999
Foundation Fieldbus Connectivity	IsaTech 1999
Plug-n-play Control Systems Integrating Foundation Fieldbus and OPC	IsaTech 1999
OPEN TECHNOLOGIES: The End of Proprietary Solutions	IsaTech 1999
Operação Remota de rede Fieldbus: telemetria via cabo, rádio e fibra ótica	Intech Brazil 1999
Advanced Control Strategies and Foundation Fieldbus - Powerful Association	ISA 2000
Optimizing Process Control Using Field Devices	ISA 2000
Configuring Foundation Fieldbus Strategies (Tutorial Session)	ISA 2000
Process Control Performance on a Foundation Fieldbus System	ISA 2000
Applying Foundation Fieldbus in a Hazardous Area	ISA 2000
Building Homogenous Systems Using the Fieldbus Programming Language and HSE	ISA 2000
A Framework for a Foundation High Speed Ethernet (HSE) Linking Device	ISA 2000
Implementing Fieldbus In Large Control Systems	ISA 2000
Vantagens e questões mais freqüentes sobre o Fieldbus baseado na arquitetura FCS.	Intech Brazil 2000
Foundation Fieldbus: a volta ao campo	Intech Brazil 2000
Safety´s easier through FF	Intech 2000
Digital bus installations are safer and cheaper	Intech 2000
Redundância High Speed Ethernet baseado em computação reconfigurável	ISA Brasil 2001
Servidor WEB em plataforma Fieldbus HSE	ISA Brasil 2001
Visão geral e perspectiva do protocolo HSE	ISA Brasil 2001
Foundation High Speed Ethernet Fieldbus	ISA 2001
Controlling your plant with Foundation Fieldbus Systems	ISA 2001
Fondation Fieldbus: A solution for problems you didn't even know you had	ISA 2001
How to deal with real applications using Foundation Fielbus	ISA 2001
Understanding control performance on a Foundation Fieldbus System	ISA 2001

Application Gallery

Smar has Fieldbus experience in practically all industry segments in all continents.



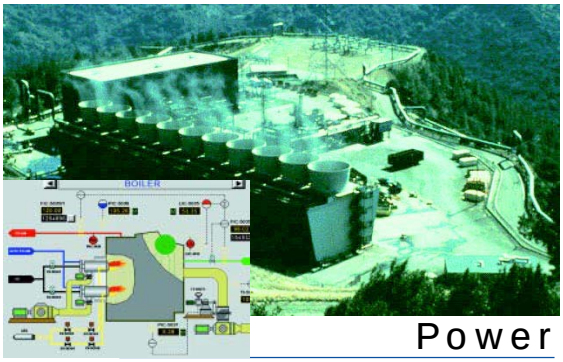
Glass

• Furnaces • Cooling



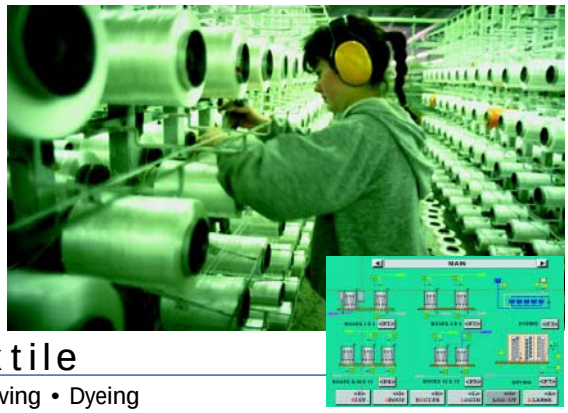
Mining

• Ore Dressing • Pyro/hydro Processing
• Electrolytic Refining • Crushers • Ball Mills • Classifiers
• Filters • Flotation Devices • Magnetic Separators
• Calcining Kilns • Digesters • Evaporators



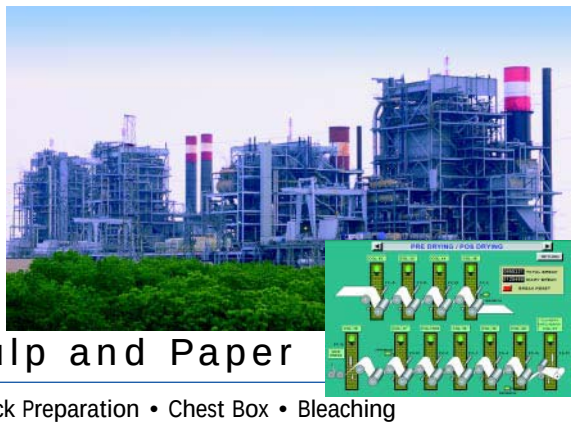
Power

• Burner Management Strategies
• Performance Monitoring • Safety Shutdown



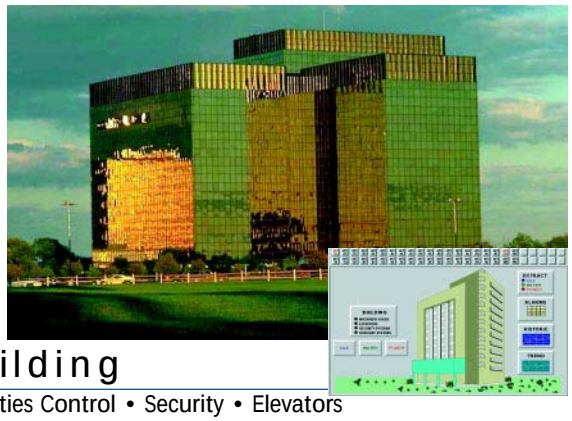
Textile

• Weaving • Dyeing



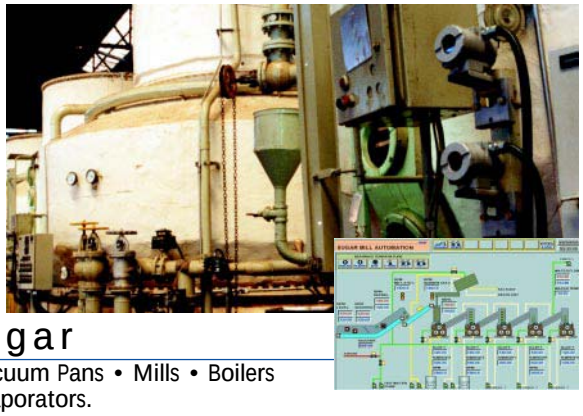
Pulp and Paper

• Stock Preparation • Chest Box • Bleaching



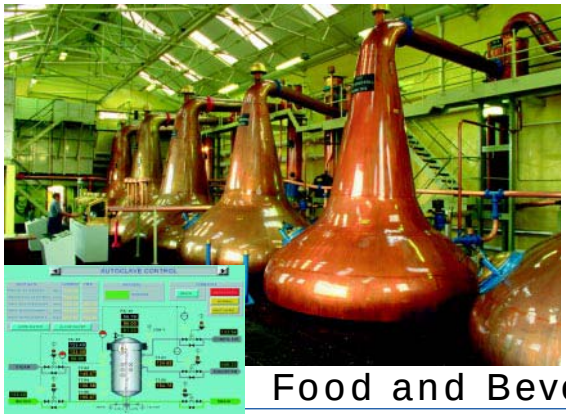
Building

• Utilities Control • Security • Elevators



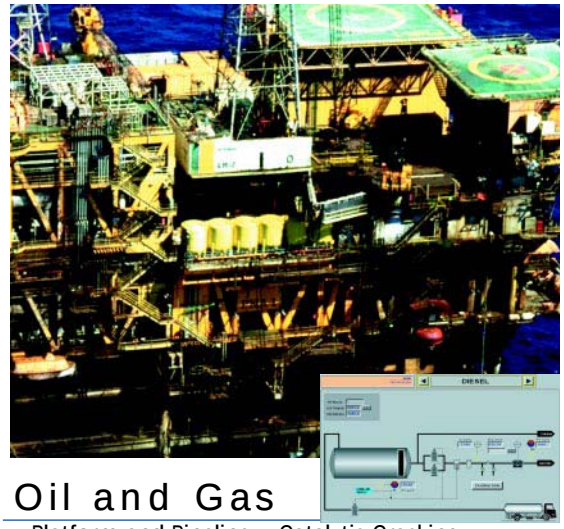
Sugar

• Vacuum Pans • Mills • Boilers
• Evaporators.



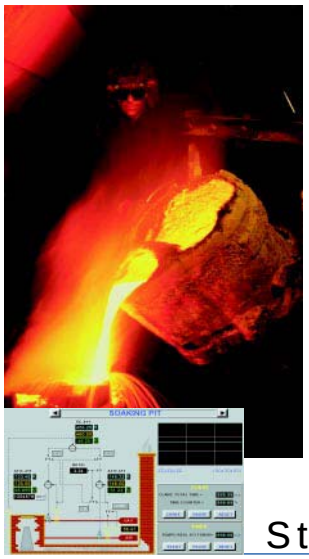
Food and Beverage

- Evaporators • Blending and Batching
- Retorts •Fermenting



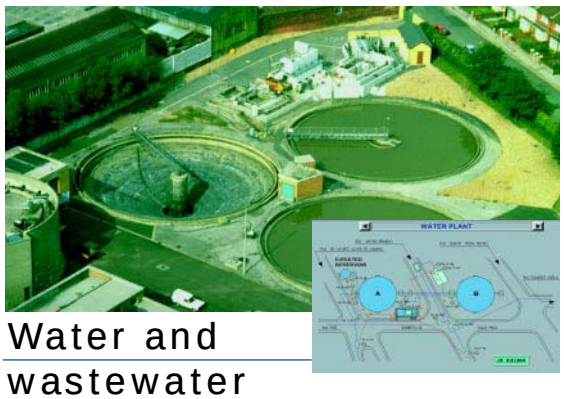
Oil and Gas

- Platform and Pipeline • Catalytic Cracking
- Distillation Columns • Blending • Reactors



Steel

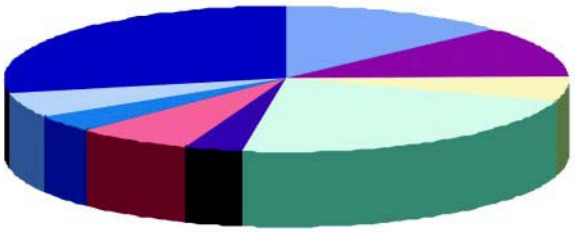
- Blast Furnaces • Continuous Casting
- Reheating Furnaces • Biological Waste Treatment
- Annealing Lines • Sintherization



Water and wastewater

- Filter Backwash • Chemical Feed • Clearwell Level • Finished Water Pumping • Remote Workstation Monitoring • Sludge Process • Sequencing Batch Reactor Plants • Rotating Biological Contactor Plant • Sludge Dewatering • Incinerator Control • Return and Waste - Activated Sludge Pumping

Applications





Expertise

Smar has thousands of applications.



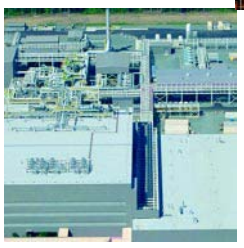
We selected some examples on the following pages.











Companhia Energética Santa Elisa



Companhia Energética Santa Elisa is one of the largest sugar and alcohol producers in the world. The plant has capacity to process 32.000 tons of sugar cane per day producing 2.000 tons of sugar and 1.400.000 liters of alcohol per day.

The automation project started in 1997, involving all the production areas in the mill and divided in three parts: the steam generation plant, the sugar manufacturing plant and alcohol distillery plant.

The SYSTEM302 allowed the distribution of the logical functions among 170 field devices integrated to 15 LC700 - Programmable Controllers with full integration between the continuous controllers and discrete variables available in the Fieldbus network. The operation is monitored in 11 workstations loaded with AIMAX man machine interface.

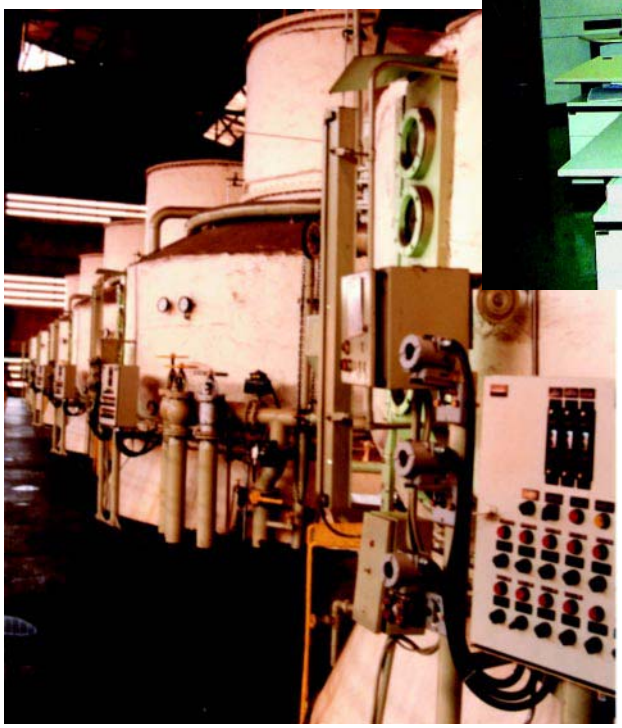
The automation project for steam generation involved control of 8 boilers. The Programmable Controllers were used for on/off commands and for security interlocking of the engines and bagasse conveyor belts.

The project for the sugar manufacturing plant used 65 field devices, 5 Programmable controllers and 5 operating stations distributed in 10 Fieldbus communication channels. All of the continuous controls and monitoring were done in the Fieldbus network. The discrete signals were handled by the programmable controllers The FB700, Fieldbus modules, on the programmable controllers provided full interaction between discrete and continuous variables.

The project for the alcohol distillery involve 2 operating stations, 3 Programmable Controllers and 67 Fieldbus devices distributed in 6 Fieldbus channels. The Fieldbus devices took care of the continuous control. The Programmable Controllers took care of the monitoring and temperature alarm handling. The PCI302 module inside the stations took care of the bus communication management and calculated on-line density providing INPM degree of the alcohol.

The system started full operation in March 1997 allowing several benefits Steam production increased, the process became much more stable and the sugar boiling process took an impressive 30% increase in productivity. Santa Elisa was the first big sugar cane mill to install a complete automation system based on Smar SYSTEM302 Fieldbus based system.





Deten



Deten Chemicals S.A. is a Linear Alkyl Benzene (LAB) producer, located in the petrochemical zone of Camaçari, Bahia, Brazil. LAB is a basic raw material used in detergents. Deten became the first in the world to install and operate a Fieldbus-based control system even before a final Fieldbus standard had been reached.

All Deten production comes from two twin plants. By the time of the Fieldbus installation the plants were 15 years old and a large part of the instrumentation was still pneumatic.

Deten chemical processing demands reliability, flexibility and long/short-term economic benefits, among critical applications as reactors, batch process, as well as boiler and compressor controls.

In December 1994, the company undertook an initial Fieldbus project to test the system in just one part of the plant, in a kind of pilot project. The upgrade firstly involved adding a process optimization unit to one of the plants, and a digital control system based on smart transmitters and multi-loop controllers from Smar was used to control the unit. There were 63 Fieldbus-compatible devices installed and 43 control loops. The field devices were divided in four Fieldbus networks. The four networks were connected to an interface card, developed by Smar that takes care of the Fieldbus communication and performs other functions. This board, called the Process Control Interface (PCI), was plugged directly into the internal 16-bit ISA bus of a computer.



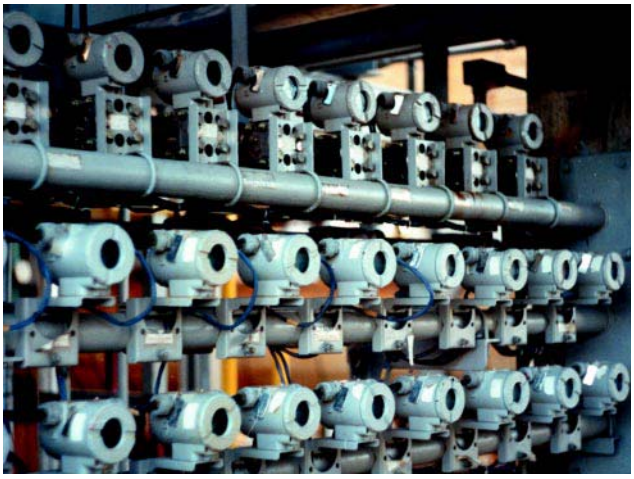
Secondly, the other plant also received an optimization unit, but this unit was implemented with the Fieldbus technology. This installation involved a total of 980 field devices which included LD302 - Pressure Transmitters, TT302 - Temperature Transmitters, FI302 - Fieldbus to Current Converters, FP302 - Fieldbus to Pressure, IF302 - Current to Fieldbus, flow devices and level measurement equipment. It also included 13 LC700 - Programmable Controllers equipped with Fieldbus network interface cards, and 12 operator workstations running supervisory software that is linked with the network.

Based on the success of running this project for more than 12 months without any problems, a larger project was undertaken in 1996 encompassing a wide range of Fieldbus instruments, including pressure and temperature transmitters, Fieldbus current converters, Fieldbus pressure converters and Programmable Controllers with Fieldbus communication modules. This project turned the plant 100% controlled by Fieldbus.

The plant is designed for full redundancy to prevent any single failure to stop the process, it has many redundancy levels allowing non-interrupt safety operation. In addition, triple-redundant devices are installed on critical control loops, and safety and status information are sent between the PLCs as a watchdog timer.

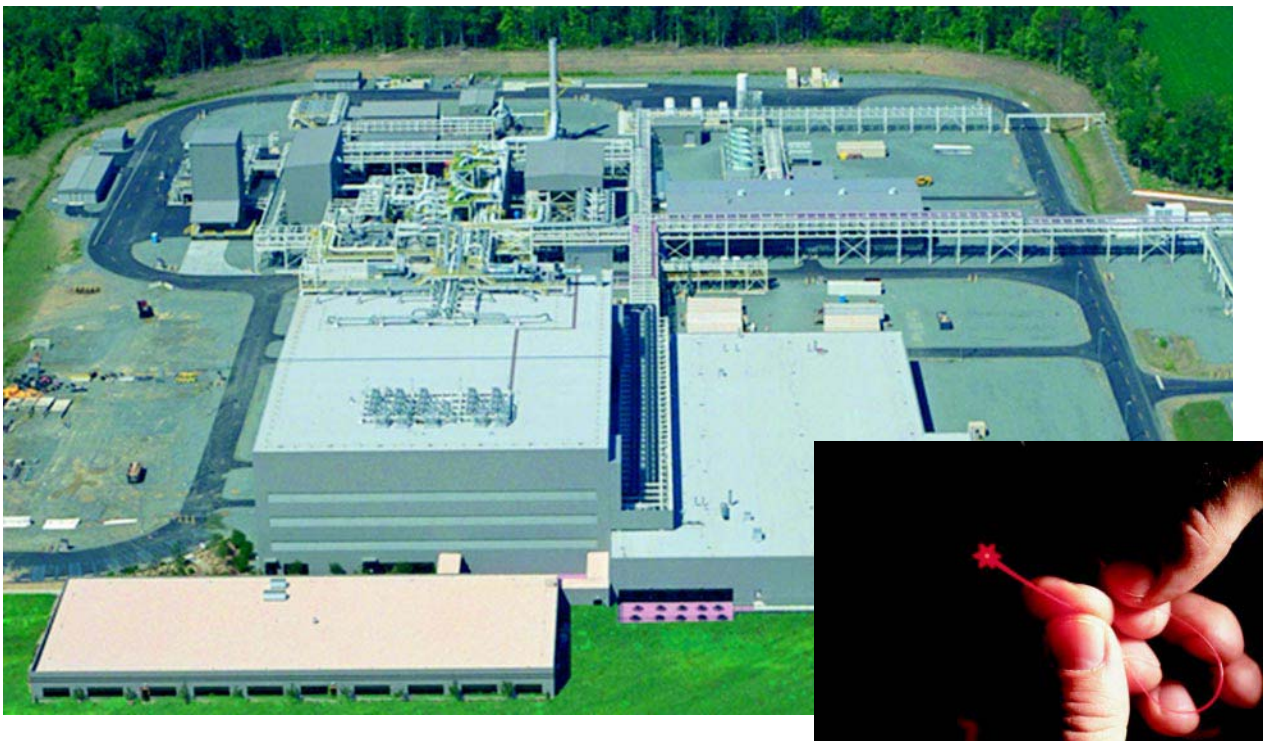
In 1997, Smar demonstrate the use of the Internet to monitor the chemical plant at the Fieldbus World Congress in São Paulo, Brazil. Smar used a dial-up connection provided by an access provider. All that was need was an account with the provider, a fax-modem card, and to configure the computer. The application basically consisted of one server computer, connected directly to the field devices, and one client computer, that was in the Smar booth. The application demonstrated features including: remote maintenance, diagnosis and operation; process integration; and how all the information that Fieldbus has to offer can be easily sent through Internet, making remote maintenance possible.

The integration of Internet technology into the Fieldbus control system improves the ability to oversee process operations. With this capability, plant personnel can verify process variables, perform on-line maintenance and monitor supervisory screens.



Reductions in design, installation and start-up costs made possible by Fieldbus have produced significant economic benefits to Deten. The company has calculated a total project saving of 32 to 45 percent. This includes savings in cable costs of approximately 97 percent. Other installation savings were achieved through reductions in cable tray and panel costs.

Corning Concord



Corning Incorporated is the world's largest producer of glass fiber for telecommunications.
In 1997, Corning decided to build a modern new facility in Midland, NC. The plant located near Charlotte, NC began producing fiber in the spring of 1999. The control system technology, chosen for this facility, was to be the most advanced in the world. Corning chose Smar SYSTEM302 to be the backbone of this new facility.
With the successful start-up in 1999, Corning's production needs continued to grow. In December 1999 Corning decided to expand the Concord facility, doubling it's capacity making it the largest plant of its' type in the world. Smar SYSTEM302 would control this new addition to the plant. With this new addition, the largest operating Foundation Fieldbus system in the world was getting larger.
The plant has over 500 Fieldbus devices (such as TT302 - Temperature Transmitters, FI302 - Fieldbus to Current Converter and IF302 - Current to Fieldbus Converter), 16 redundant servers, 24 operator stations and over 2,600 I/O discrete points running over the Fieldbus system. One unique feature of the system is that it is designed for full redundancy to prevent any single failure to stop the process. Each Fieldbus network is powered from both ends of the bus using redundant PS302 - Power Supply for Fieldbus. Also, on each end of the networks there is an interface card Smar PCI. These interface cards are in redundant I/O servers. The I/O servers have redundant Ethernet cards and send their data to redundant data servers. The next step was to include redundant Ethernet fiber switches and 16 operator consoles.
The control strategies implemented by this system include Feed back, Feed forward, Split ranges, Override, Cascade Control totally performed by the field devices.

Cognis (Formerly Henkel)



Cognis S.A, German company leader in specialty chemicals, has a plant located in Jacarei, São Paulo, that produces raw material for Pharmaceutical and Cosmetic Industry from vegetal oil. It processes 20 t/d of vegetal oil.

The SYSTEM302 is controlling the sterilization process, reactor, distillation towers, vacuum system, tanks level, and product stock. The Batch Control is performed in conjunction of the field devices and LC700 - Programmable Controller exchanging data through the Fieldbus module installed in the Programmable Controller, but that is also part of the Fieldbus network. The Programmable Controller also is responsible for the interlock system, and transportation for raw material and others products necessary for the plant.

The plant has 212 Fieldbus devices, 2 I/O servers, 3 operator's station, and over 930 I/O points. There are up to 16 devices per bus. The control strategies implemented by this system include Feed back, Feed forward, Split ranges, Override, Cascade Control totally performed by the field devices.

It is an interoperable system where Smar Field devices as LD302 - Pressure Transmitters, TT302 - Temperature Transmitters, FI302 - Fieldbus to Current Converters, FY302 - Valve Positioner and IF302 - Current to Fieldbus Converters are connected at the same bus with 19 Vortex Flowmeters from a third part supplier.

Binzhou Chemical Group



Binzhou Chemical Group (BCG), located in Binzhou, Shandong Province, one of China's major chemical producers, was formed in the late 1960's and operates several production facilities including a refinery, a chlor-alkli plant, a power plant, and a Propylene Oxide (PO) plant. These plants center around 2 main product categories: petrochemical products and specialty chemicals.

BCG previously installed Fieldbus on a boiler control project and used SYSTEM302 to control the system. This project assured that the use of Fieldbus improved diagnostic capabilities, reduced plant operating costs, minimized the effort required for the control system configuration, installation and commissioning. Because of the success on the boiler, BCG selected a Fieldbus-based solution to automate the PO plant. Unlike the boiler application, the PO plant presents a number of difficult control challenges, such as being extremely corrosive, highly toxic and highly explosive.

The Fieldbus system has met such important technical criteria to enable proactive maintenance. The features such as intrinsic safety, system redundancy, simplified safety and wiring, and remote instrument diagnostics all are part of the maintenance characteristics.

Installation of the Fieldbus-based control system was completed in November 1999. The system, which controls 22 individual control loops, includes two redundant workstations performing all operations, engineering and maintenance as well as 36 field devices spread over four Fieldbus networks. Both workstations connect the Fieldbus networks through redundant, independent interfaces. This configuration provides two paths from the control network to the operator with automatic switchover, ensuring higher system availability and fewer shutdowns.

Among the Fieldbus devices installed were LD302 - Pressure Transmitters, TT302 - Temperature Transmitters, IF302 and FI302 field-mounted converters for interfacing 4-20 mA to conventional equipment such as some analyzers. These devices have a wide diagnostic coverage giving reliable status information. SB302 - Safety Barriers with built-in repeaters were used to allow several barriers to be connected on the same network in the safe area requiring only a single interface port for a network of 12 devices.

A LC700 - Programmable Controller was used for discrete control and conventional I/O. Syscon was the client application used for configuration and maintenance of field devices and to build the control strategies. System engineering was simplified by using the host system's preconfigured templates for graphics and control strategies. BCG selected AIMAX for Windows NT for Human Machine Interface because of its OPC client and Chinese language.

Best of all, system commissioning is achieved with "plug-and-play" simplicity because field devices are automatically detected at the time they are connected.

The company's experience demonstrates that Fieldbus' robust device diagnostics improve predictive maintenance, and in doing so, minimize costly downtime. The use of standardized Fieldbus programming and function block languages simplifies training, and allows regulatory control strategies to be developed in much less time. Because Fieldbus enables regulatory control to be performed in field devices, fewer centralized controllers are required on the PO process. The technology also provides substantial reductions in I/O, and supports a smaller, more efficient supervisory control architecture.

The SYSTEM302 has been providing continuous, reliable control performance since its commissioning and start-up.

CEF Power Plant

The Comision Federal de Electricidad (CFE) is the power generation department in Mexico. It has 4 Fieldbus systems. The first Fieldbus system in Mexico started operations at the Mazatlan Thermoelectric Power Plant located in the northeast of Mexico. A high-pressure boiler producing 600 Ton/hr steam at 150 kgf/cm^2 powered this unit. The loops involved were Master Pressure control, Fuel Flow control (Crossed limits), 2 Air dampers with balance station, Oxygen trim, Drum level (3 elements) control, Feed water flow control, 1 Deaereator level control, Steam flow (as feedforward), and LC700 - Programmable Controller for interlock and alarming.

Since 1998 this system has been working continuously without any shutdowns nor problems of any kind, and 50% cost reduction compared with traditional DCS/PLC.

After a long time evaluating SYSTEM302 performance, in 1998, the plant management decided to install a second system for the unit 2 with same capacity and control loops. As expected the new SYSTEM302 installed works just like the first one, but with a new record of 1-week installation time, so far no maintenance was yet required in any of the two systems.

There is another power plant at Coahulia "Carbon II", where SYSTEM302 is handling 24 PID loops for different application in the Boiler, with plans to install a larger system for cross limits and level control. This plant also has 2 LC700 - Programmable Controller systems with more than 30 I/O cards each for several services.

In December 1999, a new SYSTEM302 was added to the list. The thermoelectric power plant at Guaymas succeeded the start up of their first SYSTEM302. The Unit 1 of this plant is also 158 000 kW. In this application the LC700 is responsible for the start and shutdown sequences for the boiler.

To date a total of 790 000 kW is being produced in Mexico using Smar SYSTEM302.



Ilustration photo

BASF

Smar participated in a partnership with Krohne/Conaut to supply equipment with the Profibus PA protocol, that included the configuration, to revamp the Resin Plant of the Paints Unit of Glasurit (BASF group), located in São Bernardo do Campo, SP. This is the largest installation with Profibus PA technology in Brazil. The Profibus PA series of Smar was launched in October of 1999 at Interkama, Dusseldorf, Germany, in response to the request of the European market, where this protocol is stronger.

A total of 109 instruments were used for the control and indication of the chemical reactors and tanks farm of Glasurit. The equipment was supplied in two stages. The first stage was started in April 2000 and the second one July 2000. The first stage was comprised of 30 Profibus PA pieces of equipment, provided by Smar as well as other suppliers.

The Profibus PA protocol was chosen by BASF based on their existing equipment that uses the Profibus DP protocol. Also the Supervisory System already had interfaces for this protocol.

The project, called FRIAS 3, included two supervisory stations operating with ABB software "DCS Advant Controller ". Two ABB Programmable Controllers, each one with a communication module Profibus DP, make two Profibus DP networks. A total of 21 DP/PA Couplers of the Pepperl & Fuchs were connected, totaling 21 Profibus PA buses.

There are 76 Smar devices distributed in these buses: 20 Pressure Transmitters LD303/LD293, 30 Temperature Transmitters TT303, 25 Valve Positioners FY303 and 1 Current to Profibus Converter IF303.

A configuration and maintenance station with Siemens CP5611 module running the Siemens PDM configurator, was part of the project and was used by Smar to configure the Profibus PA buses.

All of control strategies of the system were developed by ABB and uploaded to its Programmable Controllers. Smar supplied the files necessary for these strategies, such as the GSD_File (Device Master Date) and others.



Ilustration photo

Petrobras



Petrobras photo library

The world's first Foundation Fieldbus System on an Off Shore application is situated in Campos Basis area known as "Namorado 1" Platform in Rio de Janeiro, Brazil. Petrobras, a Brazilian oil company and one of the twenty major oil companies in the world producing more than 1.6 million of barrels per day, operates this Platform.

The installed System measures Natural Gas Pressure and Flow. It operates with a network connecting the Field Devices and the operating stations in a control room. This platform is responsible to control the gas pipelines from and to Roncador, Marlim-S, Cabiunas, Marlim PCH-1 Namorado 2.

Fieldbus technology was chosen due to its easy integration of all data generated by the system instruments, and its modern architecture, with distributed control located inside field devices.

As apart from providing much more information very fast and most reliably, it also brings down costs of cables, cable trays, labor for wiring and trays and finally maintenance. The time spent in installations is also significantly reduced.

The project includes 21 devices and 6 DFI302 - Fieldbus Universal Bridges in redundancy, monitored centrally by Intouch HMI software. In the Control Room there is a provision for two Operator Stations and two Engineering Stations connected through an Ethernet fiber optic network.

Gas-line pressure control is based on a valve and two transmitters, one upstream and the other downstream of the valve. Two PIDs are linked to an input signal selector function block that provides the output for the control valve. Two control loops manage the plant's deaeration system for the water injection. The architecture is designed to have all Fieldbus gateway nodes installed on pressurized panels in the field. These then connect to the control room via fiber optics.

Smar also achieved the first commercial installation in the world of HSE (High Speed Ethernet) technology using SYSTEM302 on this platform, which is also the first installation in the world on a platform of H1 Fieldbus.

Smar coordinated the project and also supplied all Field Devices, Control Hardware and Software.

US Navy

Smar has deployed its SYSTEM302 to sea. Since 1998 Smar has been supplying control systems and instruments to the U.S. Navy. Nowadays several ships are already using SYSTEM302 to control their fuel injection system, their engine lubrication oil injection system, as well as the boiler pressure reduction loops.

Smar's SYSTEM302 is not only improving the control loop performance, but also delivering fault tolerant installations using Fieldbus status and diagnostic information. The fact that all of the loop controls can be done inside the field devices, even when all of the control systems are down, was one of the key features for the acceptance of a Fieldbus system in the U.S. Navy. SYSTEM302 proved to be an extremely reliable system, resistant to vibrations, high temperatures and strong electromagnetic interference of extreme conditions.

Other characteristics of a true Fieldbus System like SYSTEM302 were extremely useful in the ship. Cable reduction, less equipment in the control room, control in the field, instantaneous status message on failures, were some of the features analyzed by the U.S. Navy.

Today, Smar is proud to say that we have our equipment being used by one the most demanding customers in the world, the U.S. Navy, where high reliability, high availability and high performance are a must.



Reference List

Customer	Segment
Cotrefal	Agriculture
Dedini - S.J. da Boa Vista	Alcohol
Usina Cerradinho	Alcohol
Usina Cerradinho	Alcohol
Usina S. João de Araras	Alcohol
Açucareira Corona	Alcohol & Sugar
Açucareira Corona	Alcohol & Sugar
Cia. Energética Sta.Elisa	Alcohol & Sugar
Cirio Brasil	Alcohol & Sugar
Coop. Agropec. Rolandia	Alcohol & Sugar
Coopernavi	Alcohol & Sugar
Coopernavi	Alcohol & Sugar
FB Açúcar e Álcool	Alcohol & Sugar
Ibate SA	Alcohol & Sugar
Ind. Açucareira S. Frco.	Alcohol & Sugar
Irmãos Franceschi	Alcohol & Sugar
Lwarcel - Celulose e Papel	Alcohol & Sugar
S/A Usina Coruripe Açúcar e Alcool	Alcohol & Sugar
Us. Açucareira Jaboticacal	Alcohol & Sugar
Us. de Açúcar Sta.Terezinha	Alcohol & Sugar
Usina Alto Alegre	Alcohol & Sugar
Usina Alto Alegre	Alcohol & Sugar
Usina Caete	Alcohol & Sugar
Usina Cerradinho	Alcohol & Sugar
Usina Maracai	Alcohol & Sugar
Usina Maracai	Alcohol & Sugar
Mercedes Benz	Automotive
Antartica	Beer Industry
Nordon	Beer Industry
Primo Schincariol	Beer Industry
Cervejaria Ref. Convenção	Beer Industry
Palmol	Boiler
Ref. Milho Brasil	Boiler
Refinería Cienfuegos	Boiler
Aços Villares	Boiler
Seattle Steam	Boiler
Chemin	Building automation
Construtora Pessoa	Building automation
Cementos Catatumbo	Cement
Cementos Lima	Cement
ICAL	Cement
Furnas	Cement
Anling Chemical	Chemical
Binzhou Chemical	Chemical
Binzhou Chemical	Chemical
China National Chemical	Chemical
Cia. Nitro Quimica Bras.	Chemical
Clorox/ CICA	Chemical
Cognis	Chemical
Cognis Brasil Ltda	Chemical
Deten	Chemical
Deten	Chemical
Deten	Chemical
Deten	Chemical
Fermic/JGL	Chemical
Fermic/SICA	Chemical
Fertilizer Plant, Han Steel Group	Chemical
Frunot SA	Chemical
Frunot SA	Chemical

Reference List

Customer	Segment
Henkel	Chemical
Henkel	Chemical
Henkel	Chemical
Jilin	Chemical
Kodak Brasileira Com. Ind. Ltda	Chemical
Kolynos	Chemical
Polioles S.A	Chemical
Primax	Chemical
Productos Quimicos y Anexos Proquimsa S.A	Chemical
Rhodia	Chemical
Rhodia	Chemical
Rhodia	Chemical
Rhodia - Ster Fibras e Resinas Ltda	Chemical
Rhodia Brasil Ltda	Chemical
Rhodia Brasil Ltda	Chemical
Sapici	Chemical
Seattle Steam	Chemical
ACELAB	Chemical
Binzhou Chemical	Chemical
Controlflux	Chemical
Elf - Penwalt/Cobando	Chemical
Penta/ SICA	Chemical
Quimica Central/JGL	Chemical
Southern Edison	Chemical
Triken	Chemical
Triken	Chemical
Usine D'Amiens	Chemical
Kolynos do Brazil	Chemical & Petrochemical
Petrobras - Petroleo Brasileiro	Chemical & Petrochemical
Petrobras - Petroleo Brasileiro	Chemical & Petrochemical
Rhodia	Chemical & Petrochemical
Rhodia	Chemical & Petrochemical
Rhodia Brasil	Chemical & Petrochemical
Shanghai Dongdao Carbon Chemical Industry	Chemical & Petrochemical
Univen Petroquimica	Chemical & Petrochemical
Fundação de Pesquisa e Assess.	Demo/Training
HV & L	Demo/Training
Martin Controls	Demo/Training
OY E Sarlin AB	Demo/Training
OY E Sarlin AB	Demo/Training
OY E Sarlin AB	Demo/Training
Standard Auto	Demo/Training
Asflow Control	Demo/Trainning
Bem - Gurion Univ.	Demo/Trainning
BRCS Instrumentation	Demo/Trainning
Chemtech	Demo/Trainning
CIED	Demo/Trainning
Controlflux	Demo/Trainning
Desin	Demo/Trainning
Eleinca	Demo/Trainning
Field Instr. Control	Demo/Trainning
Field Instr. Control	Demo/Trainning
Fundação CSN	Demo/Trainning
Fundação CSN	Demo/Trainning
Fundação CSN	Demo/Trainning
GSE	Demo/Trainning
GSI	Demo/Trainning
IBP	Demo/Trainning
ISA	Demo/Trainning

Reference List

Customer	Segment
JGL Controls	Demo/Trainning
JGL Controls	Demo/Trainning
KDG	Demo/Trainning
Konics Co. Ltd	Demo/Trainning
Lambton College	Demo/Trainning
Mitsubishi Electric Corporation	Demo/Trainning
Petrobras - Petroleo Brasileiro	Demo/Trainning
Petro - Tech	Demo/Trainning
Provence Automation	Demo/Trainning
SENAI	Demo/Trainning
SENAI	Demo/Trainning
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SENAI	Demo/Trainning
SENAI	Demo/Trainning
SENAI	Demo/trainning
Singapore Polytechnic	Demo/Trainning
Singapore Polytechnic	Demo/Trainning
Statoil	Demo/Trainning
Technical Specialties	Demo/Trainning
TECSUP	Demo/Trainning
Tillquist	Demo/Trainning
Tillquist	Demo/Trainning
Tillquist	Demo/Trainning
Trondheim University	Demo/Trainning
Univ. Estadual Paulista	Demo/Trainning
Univ. Católica	Demo/Trainning
Univ. Católica	Demo/Trainning
Univ. de Carabobo	Demo/Trainning
Univ. Viçosa	Demo/Trainning
Univ.Fed.Minas Gerais	Demo/Trainning
University Rio Cuerto	Demo/Trainning
Tequila Cuervo/J. Diaz	Destilery
Equipalcool	Equipment
Quimpac S.A.	Equipment
PDVSA / INTERVEP	Flow Station
Alicorp S.A. Planta: Fideería	Food
Cervejaria Ref. Convenção	Food & Beverage
Cervejarias Kaiser	Food & Beverage
COCAMAR	Food & Beverage
COCAMAR Oleo Agro	Food & Beverage
Comercio e Industria Brasileira	Food & Beverage
COMIGO S.A	Food & Beverage
Corn Products Brasil	Food & Beverage
Corn Products Brazil	Food & Beverage
Corn Products Brasil	Food & Beverage
Eleinca C.A.	Food & Beverage
Firmenich	Food & Beverage
Frilli Dis S.Tommaso	Food & Beverage
Frilli - La Versa	Food & Beverage
INALCA	Food & Beverage
Marine Lipids	Food & Beverage
Nabisco	Food & Beverage
Nabisco	Food & Beverage
Nestle	Food & Beverage
Nestle - Forima	Food & Beverage
Nuova Campari	Food & Beverage
Ref. Milho Brasil	Food & Beverage

Reference List

Customer	Segment
Ref. Milho Brasil	Food & Beverage
Resende Alimentos	Food & Beverage
Resende Alimentos	Food & Beverage
Sadia Alimentos S/A	Food & Beverage
Cebrace - Cristal Plano Ltda	Glass
Cebrace - Cristal Plano Ltda	Glass
Corning Glass	Glass
Corning Incorporated	Glass
Corning Incorporated	Glass
ALBRÁS	Metallurgy
ALBRÁS	Metallurgy
ALBRÁS	Metallurgy
ALBRÁS - Aluminio Brasileiro	Metallurgy
FURNAS	Metallurgy
Gerdau	Metallurgy
Mannesmanm	Metallurgy
Borden Chemical	Methanol Cond. Control
BIPC	Methanol Process Lines
Minera Michilla	Minerals
Minera El Tesoro	Mining
Minera El Tesoro	Mining
Minera El Tesoro	Mining
Muis Trading Company	Mining
Firmenich	Multipurpose Plant
CUPET	Oil & Gas
CUPET	Oil & Gas
CUPET	Oil & Gas
CUPET	Oil & Gas
CUPET	Oil & Gas
EPEP - Occidente	Oil & Gas
Liquigás	Oil & Gas
Minasgas	Oil & Gas
Petrobras - Petroleo Brasileiro	Oil & Gas
Petrobras - Petroleo Brasileiro	Oil & Gas
Petrobras - Petroleo Brasileiro	Oil & Gas
YPF/Esco	Oil & Gas
Cupet	Oil Refinery
Cupet	Oil Refinery
J. Diaz	Oil Refinery
Jaruco	Oil Refinery
Jaruco	Oil Refinery
PEMEX	Oil Refinery
Pemex/Controlflux	Oil Refinery
Refinería Hnos. Díaz	Oil Refinery
Total - Weisz/Esco	Oil Refinery
UNIVEN	Oil Refinery
UNIVEN	Oil Refinery
YPF	Oil Refinery
Belgo Mineira	Oven System
White Martins	Oxigen Injection
Tintas Renner	Painting Products Plant
Daishowa America	Paper Mill-Training
BIPC	Petrochemical
Fox Petroli	Petrochemical
Fox Petroli	Petrochemical
Fox Petroli	Petrochemical
Petrobras - Petroleo Brasileiro	Petrochemical
Petrobras - Petroleo Brasileiro	Petrochemical
Petrobras - Petroleo Brasileiro	Petrochemical

Reference List

Customer	Segment
Petrobras - Petroleo Brasileiro	Petrochemical
Petrobras - Petroleo Brasileiro	Petrochemical
Petrobras - Petroleo Brasileiro	Petrochemical
Petrobras - Petroleo Brasileiro	Petrochemical
Petrobras - Petroleo Brasileiro	Petrochemical
Petrochemical Commercial CO.	Petrochemical
Petróleo Ipiranga	Petrochemical
Petroquimica União	Petrochemical
Statoil	Petrochemical
Univen Petroquimica	Petrochemical
Univen Petroquimica Ltda	Petrochemical
Johnson & Johnson	Pharmaceutical
Smith Kline/JGL	Pharmaceutical
Seattle Steam	Power
CFE	Power Plant
CFE	Power Plant
CFE - Mexico/J. Diaz	Power Plant
International Atomic Energy Agency	Power plant
Mohave Generation	Power plant
CFE - Mexico	Power Plant
CFE - Mexico	Power Plant
Celulose Cambará	Pulp & Paper
Celulose Cambará	Pulp & Paper
Celulose Cambará	Pulp & Paper
Celulose Cambará	Pulp & Paper
Celulose Cambará	Pulp & Paper
Celulose Cambará	Pulp & Paper
Klabin	Pulp & Paper
Korsnas	Pulp & Paper
Lwarcel Celulose e Papel	Pulp & Paper
Lwarcel Celulose e Papel	Pulp & Paper
Nan Leer/JGL	Pulp & Paper
Riocell	Pulp & Paper
Selecta	Pulp & Paper
Selecta/Laplace	Pulp & Paper
Kemira Chemicals	Reactor
SAPICI	Reactor Batch Control
Firestone	Rubber
Spirax Sarco	Sales/Training
ABS	Solution Provider
Singapore Polytechnic	Solution Provider
Singapore Polytechnic	Solution Provider
Sulivan Park	Solution Provider
Sulivan Park	Solution Provider
E+H	Solution Provider
E+H	Solution Provider
E+H	Solution Provider
E+H	Solution Provider
E+H	Solution Provider
E+H	Solution Provider
E+H	Solution Provider
E+H	Solution Provider
E+H	Solution Provider
Metso Positioner Manufact.	Solution Provider
Metso Positioner Manufact.	Solution Provider
Metso Positioner Manufact.	Solution Provider
Nestle - Forima	Steam and Glycol units
Açoforja	Steel Mill
Aços Villares	Steel Mill

Reference List

Customer	Segment
Aços Villares	Steel Mill
CMM	Steel Mill
Sid. Barra Mansa	Steel Mill
Forjacero/Controlflux	Steel Plant
Mannesmanm	Steel Plant
Mannesmanm	Steel Plant
Açuc. Vale do Rosario	Sugar Plant
Açucareira Carolo	Sugar Plant
Açucareira Carolo	Sugar Plant
Açucareira Guaraní	Sugar Plant
Azucareira Teran	Sugar Plant
Azucareira Teran	Sugar Plant
Azucareira Toilman	Sugar Plant
Azucarera del Noroeste	Sugar Plant
Copersucar	Sugar Plant
FB - Açúcar e Álcool	Sugar Plant
Ind. Porto Rico	Sugar Plant
Ingenio San Carlos	Sugar Plant
Molipasa/Eleinca	Sugar Plant
Molipasa/Eleinca	Sugar Plant
Refinadora Catarinense	Sugar Plant
Refinería Cienfuegos	Sugar Plant
Usina Albertina	Sugar Plant
Usina Caeté	Sugar Plant
Usina Corona	Sugar Plant
Usina Corona	Sugar Plant
Usina Furlan	Sugar Plant
Usina Maracaí	Sugar Plant
Usina Maracaí	Sugar Plant
Usina Moema	Sugar Plant
Usina Santa Elisa	Sugar Plant
Usina Santa Elisa	Sugar Plant
Usina São Carlos	Sugar Plant
Pablo Noriega	Sugar Plant -Boiler
Usina Costa Pinto	Sugar Plant -Boiler
Azucareira Paraguaya	Sugar Process Lines
Refinería Cienfuegos	Tank Farm
Transultra-Solvay	Tank Farm
Alpargatas Santista	Textyle Industry
Fermic	Tower Recovering
Ngee Ann Polytechnic	University/School
Ana Maria Frattini	University/School
Cebrace - Cristal Plano Ltda	University/School
Centro Federal de Educação	University/School
Centro Movel Treinamento	University/School
Chamber of Commerce and Industry	University/School
Instituto Superior no Estatal TECSUP	University/School
Lee College	University/School
SENAI	University/School
SENAI	University/School
SENAI - Centro Téc. Meta Mecânica Euvaldo Lodi	University/School
Trainning Mobile Center	University/School
University of Canterbury	University/School
TERMOLYSE	Waste Recycling
Cia. Saneamento de Minas Gerais	Water Treatment
Lotil Construções	Water Treatment
Duratex Mad. Ind.	Wood



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